ACCESSIBILITY

Process and prerequisites for product testing

Version 1.1

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TOOLS FOR ACCESSIBILITY TESTING

ACCESSIBILITY TOOLS

Use a variety of tools to test accessibility.

Testing Tool / Method	Purpose / What It Does
✅ **Automated Scanning Tool**	Scans pages for predicted accessibility issues. Note: does **not** replace manual testing.
🗣️ **Screen Reader**	Reads out content from or about the page to assist visually impaired users.
⌨️ **Keyboard Navigation**	Ensures all content is accessible and usable without a mouse.
📱 **Touchscreen Interaction**	Validates content is accessible for users on touch devices.
🔧 **Element Inspector Tool**	Allows you to inspect individual page elements for accessible markup and structure.
🎨 **Contrast Checker**	Tests the contrast ratio between text and background to ensure visibility for low-vision users.

TESTING GUIDES & KEY POINTS

Automated scans

Objectives of testing

Ensure:

- Non-visual content meets accessibility standards.
- **Re-scan** after any page change, including when:
  - a modal opens
  - a dropdown item is selected
  - links or buttons are enabled/disabled

TIP: Dock the dev tools bar at the bottom of the screen (recommended).

- **Minimizes** page changes and keeps desktop view.

Screen Reader

Objectives of testing

Ensure:

- Page content is announced in a logical order.
- Provide alt text for images required to operate the page.
- **Announce identifiable** information (full names, units, categories).

Examples:

Not sufficient: "Column 5 of 9". Sufficient: "Column 5 of 9 Academic Arts High School".

Not sufficient: "Remove widget". Sufficient: "Remove notification widget".

Not sufficient: "10.5" or "3". Sufficient: "10.5 percent" and "3 percent".

Keyboard-only

Objectives of testing

Ensure:

- All mouse functionality is operable with the keyboard.

Focus indicator:

- moves logically
- **tabs** only to interactive elements
- does not get trapped on elements.

TIP: Use the ANDI bookmarklet (next slide) to view tab order. Click the "tab order" link.

TESTING GUIDES: KEY POINTS

Touch Screen	ANDI Bookmarklet	WebAIM Contrast Checker
Testing objectives **Verify:** ○ **Switch** between input methods without disrupting workflow. ○ **Example**: Use touch to open a dropdown, then use the keyboard to select an option. Focus indicator and workflow should remain consistent.	Testing objectives **Verify:** ○ **Confirm** accessible markup and structure. Select the content type, then hover to inspect elements. TIP: For issues at 200% zoom, enable the "Minimode" in ANDI to reduce the menu footprint.	Testing objectives **Verify:** **Meets** accessibility contrast standards. Enter background/foreground HEX codes or use the eyedropper to sample elements.

WHY MANUAL TESTING MATTERS

Automated Testing CAN Predict ARIA attributes, keyboard tab stops, and structural issues.

But, Automated testing can't--

Interpret context

Determine whether content is decorative or essential

Assess whether text is meaningful and understandable

Assess whether abbreviations/unique words are pronounced correctly

Test concurrent input methods

Switch between input methods (for example, touchscreen and keyboard) during a workflow

Identify and associate equivalent controls

Determine whether users can complete the same task in more than one way on a page

ACCESSIBILITY TESTING

FOR A PRODUCT

PREPARATION & EXPECTATIONS

ASSESS NEEDS

<u>Before work begins…</u>

- Complete and provide the form below with the expected conformance level and supporting details for accessibility testers.
  - Accessibility Form - Bringing a Product Up to Standard

PURPOSE: Focus testing on applicable WCAG Success Criteria.

- Tell testers whether testing supports a Voluntary Product Accessibility Template (VPAT) or maintenance/regression testing.

PURPOSE: This informs the timeline and testing approach.

ESTIMATE SCOPE

Preparation and readiness	Provide testers access to the product.
	Provide an up-to-date recorded product walkthrough (live or documented).
	Give testers at least one week to review the product and targeted workflows.
Rationale	This helps testers build an accurate, realistic timeline.
	Product familiarity helps testers plan and execute testing efficiently.

ESTIMATE WORK SCOPE

<u>What helps set the timeline?</u>

- Was accessibility considered in the build?
- How many pages are in the product?
- How many page states exist? (enabled/disabled elements, returned states, etc)
- When do modals trigger? (warnings, alerts, input fields, etc)
  - Some modals appear only in specific scenarios and are easy to miss.
    This ensures every component is tested and accounted for.
- What components does the product include?
  - Ex: Simple content—text and images only (articles/informative content.)
  - Ex: Complex content—input forms, charts, calendar pop-outs, grids, timed content, etc.
- Does the product include third-party components? (Exempt from testing)
- What documentation can the site generate? (PDFs, CSVs, etc)
- How many places can generate documentation?

PLAN THE WORK

Align on the most efficient testing approach:

A. Assign each tester one testing type; everyone tests the same page.

B. Assign each tester a different page; each uses all testing types.

C. Choose what works; brainstorm a feasible plan.

Pros and Cons:

- Option A covers WCAG Success Criteria faster but slows product coverage.
- Option B covers the product faster but slows test results.
- Overall: Option A is more thorough, organized, and efficient.
- Overall: Option B is less organized and less time efficient but saturates the product faster.

Accessibility
product testing

WHAT'S INVOLVED?

COMPLEX TESTING

- Test all pages at 200% and 400% zoom
- Test with at least one screen reader
  - Use multiple screen readers; results can vary
- Test using keyboard-only navigation
- Run tests in every page state (enabled/disabled content, open modals, return states, etc.)
- Test all site-generated documents (PDFs, CSVs, etc.)
- Test across multiple browsers

TEAM EFFORTS

- Testers and developers should review W3C resources (see Appendix C) to align on fixes.
  - Some fixes are context-specific, and some WCAG criteria include exceptions.
- Some issues require new or restructured components to meet accessibility requirements.
- Train team members and support testing.

Impact

- Shared source material improves documentation and communication, reduces rework, and increases time to address issues.

COMMUNICATING FINDINGS

Tracking Findings

- Spreadsheets
- Guiding documents
  - Baseline Accessibility Testing Guide for Product Owners
  - Non-POs can use this document as a framework for page-by-page testing

Tickets

- Link relevant WCAG Success Criteria (use the "Understanding" docs; see Appendix C: Helpful Accessibility Resources)
- Link other resources for fixing the issue
  - W3C pattern and practices guides (see Appendix C: Helpful Accessibility Resources)

Sprints

- Product owners (POs) create an Epic/Parent Issue for related tickets
  - Include Appendix C resources in every Epic description
- POs and Engineering Managers align on next steps based on:
  - Team capacity
  - Work size and complexity
  - This work and other suite priorities

TICKET COMPLETION FLOW

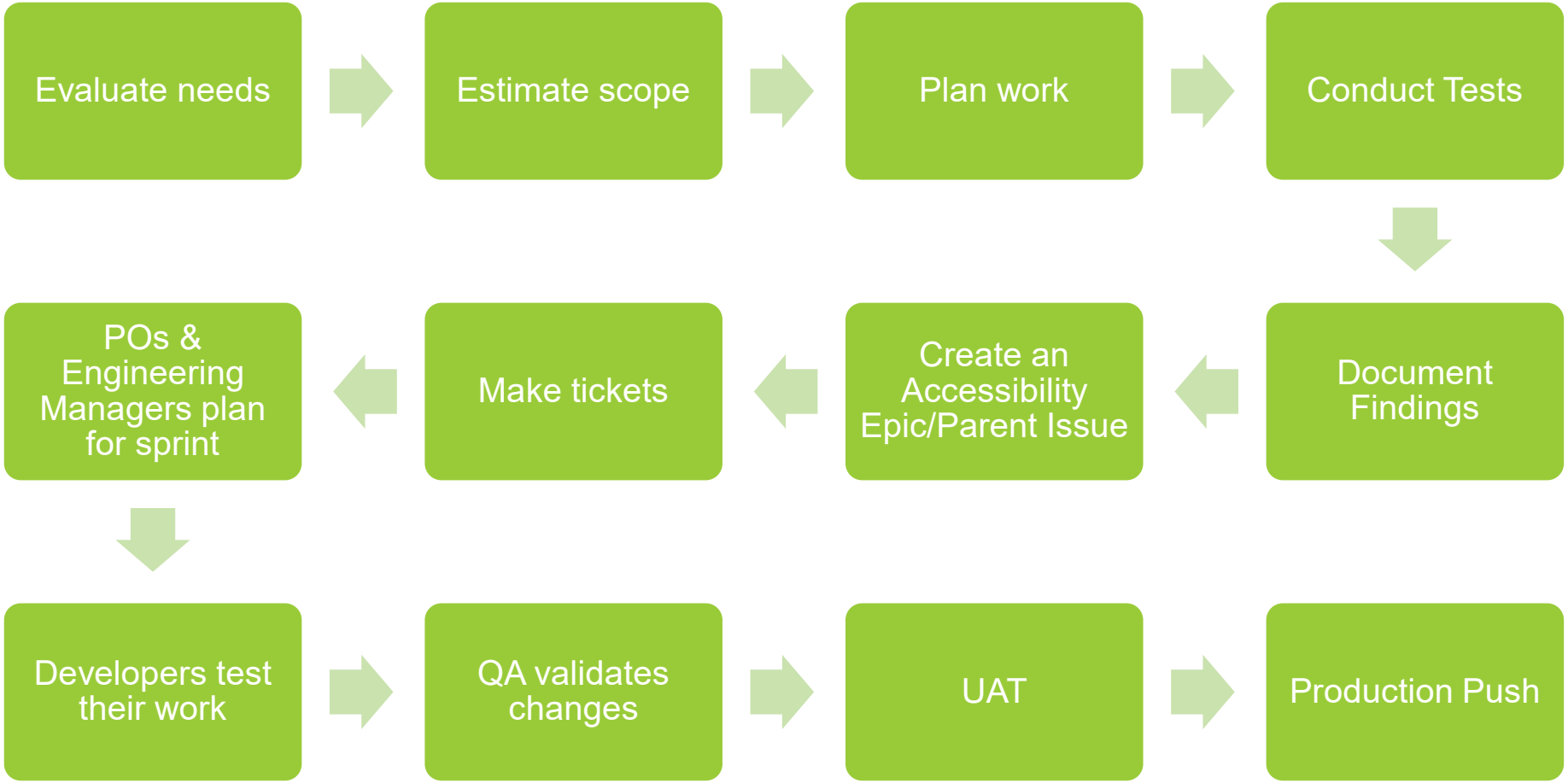
ENGINEERS USE TOOLS TO CONFIRM CHANGES WORK.

QA REVIEWS TICKETS AFTER ENGINEERING COMPLETES WORK.

PRODUCT OWNERS (POS) CREATE CUSTOMER TESTING GUIDANCE WHEN NEEDED.

UAT ALIGNS WITH SPRINT AND CUSTOMER TIMELINES.

HIGH LEVEL PLAY-BY-PLAY



ACCESSIBILITY TESTING FOR

VOLUNTARY PRODUCT ACCESSIBILITY TEMPLATE (VPAT)

PREPARATION & EXPECTATIONS

VPAT
testing

So, what is a VPAT?

VPAT OVERVIEW

A Voluntary Product Accessibility Template (VPAT) is a standardized form vendors use to document how an information and communications technology (ICT) product or service conforms to accessibility requirements.
It most commonly supports Section 508 of the Rehabilitation Act (29 U.S.C. § 794(d)).
It helps federal contracting officials and other buyers assess accessibility features and identify gaps.

Review the purpose, key objectives, and benefits of a VPAT in our What is a VPAT? document.

Evaluate the Needs

Before starting a VPAT, evaluate the product's needs and its accessibility status.

Use this form to consolidate the information needed to conduct a VPAT: Accessibility Form - If you need a VPAT

ESTIMATE WORK SCOPE

- Same preparation roadmap as general testing:
  - Know the expected conformance level
  - Receive training and reference materials
  - Familiarize yourself with the product
  - Document product details to build the timeline and approach

PLAN WORK

- Same approach as general testing:
  - Team selects the best method for their situation.
  - Prioritize confidence and comfort.

ACCESSIBILITY TESTING FOR A VOLUNTARY PRODUCT ACCESSIBILITY TEMPLATE (VPAT) SCOPE AND PROCESS

VPAT
testing

CONDUCTING A VPAT

- **Understand the expected testing standard**
  - Multiple VPAT types and WCAG conformance levels exist
- **Refer to WCAG Success Criterion**
  - The evaluator reviews applicable Success Criterion and confirms they are met on every page and in every state of the product.
- **Test full pages and processes**
  - Test full pages and complete processes using only accessibility-supported methods.
    View WCAG 2.0 Conformance Requirements for details.
- **Document findings**
  - Read VPAT instructions to understand how response terms are interpreted.
- **Use the latest VPAT template:**
  https://www.itic.org/policy/accessibility/vpat

KEY REMINDERS

Perspective matters

- VPAT is not a checklist
  - Meeting a success criterion once does not mean you can check it off and move on.

As evaluators, test each criterion against *every page, every component, in every state*.

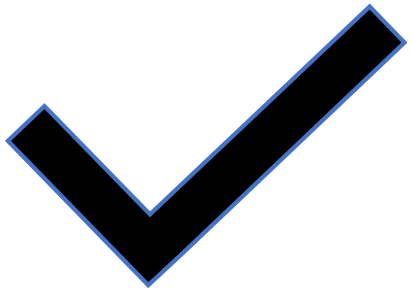
- To mark a Success Criterion as "fully supported," at least one method must pass without failure across the *entirety* of all product pages and processes; **and** when that is not possible, provide an accessible alternative that meets the same purpose.

- VPAT does not:

  List inaccessible parts of the product.

  It states which *accessibility criteria* are supported or not.

VPAT is not ideal for tracking accessibility gaps during remediation. Use spreadsheets or guided documents instead.



APPENDIX

ACCESSIBILITY TOOLBOX

Automated Scanners

- Axe DevTools
- WAVE extension
- Accessibility Insights
- Siteimprove Accessibility Checker

Code Checkers for Developers

- Axe DevTools Linter
- jsx-a11y-lint
- jest-aXe

Screen Readers

- NVDA screen reader
- Your computer's built-in screen reader

Element Inspection Tools

- ANDI bookmarklet (multi-functional)
- WebAIM Contrast Checker (contrast only)

APPENDIX C: ACCESSIBILITY RESOURCES

Quick references

Fundamental concepts

- Read me first: Why ARIA matters
- Build accessible components
- Implement ARIA practices
- Accessible names by ARIA role
- ARIA roles (definitions)

WCAG 2.1

- WCAG 2.1 criteria (all)
- WCAG 2.1 AA criteria only
- WCAG 2.1 Understanding docs

WCAG 2.2

- WCAG 2.2 criteria (all)
- WCAG 2.2 AA criteria only
- WCAG 2.2 Understanding docs

W3C resources

- W3C understanding techniques

Recommended resource

W3C Digital Accessibility Foundations (free course)

THANK YOU